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2. Hydrogen peroxide, H_2O_2, has anti-fungal properties and can be used to preserve milk. Before the milk is used to make cheese the H_2O_2 needs to be removed.

Hydrogen peroxide decomposes to form water and oxygen gas as shown below.

$$\text{hydrogen peroxide} \longrightarrow \text{water} + \text{oxygen}$$

(a) Complete the balanced **symbol** equation for the reaction. [2]

$$2\,H_2O_2 \longrightarrow \boxed{}\ \text{..............................} + O_2$$

(b) This reaction is very slow at room temperature.

A student suggests that the rate of reaction can be increased by adding a catalyst. State **one other** way that the rate of reaction could be increased. [1]

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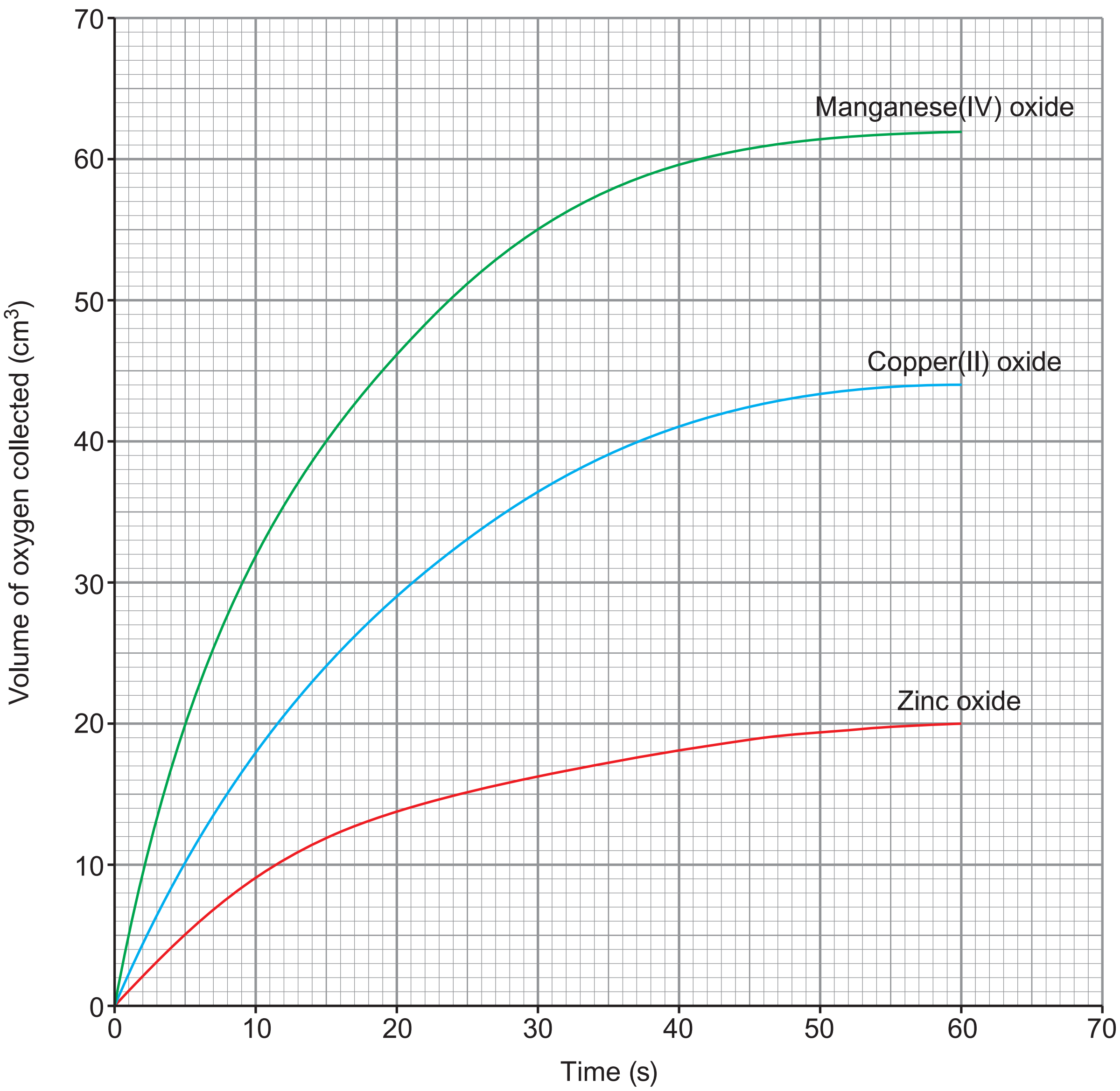
Examiner only

(c) Students investigated the decomposition of hydrogen peroxide using three different catalysts. The catalysts used were manganese(IV) oxide, copper(II) oxide and zinc oxide. Their results are shown on the graph below.



Use the graph to answer the following questions.

(i) State the volume of oxygen collected in 10 seconds when manganese(IV) oxide is used. [1]

Volume = ... cm^3

(ii) State the time taken to collect $20\,cm^3$ of oxygen when copper(II) oxide is used. [1]

Time = ... s

3445U301
05

Examiner only

(iii) One student says that zinc oxide is the best catalyst to use.
Explain whether the student is correct. [2]

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(d) The mass of catalyst used at the start of each experiment was 1.2 g.
State the mass of catalyst at the end of each experiment. [1]

Mass = ... g

(e) Milk used in cheese making is stored in the refrigerator at 4°C. Room temperature in the kitchen is 24°C.

(i) Calculate the difference in temperature between the refrigerator and the kitchen. [1]

Temperature difference = ... °C

(ii) It is claimed that for every temperature decrease of 10°C, the milk will take twice as long to go sour. The milk sours in 10 hours when left out of the refrigerator in the kitchen. Calculate the time it takes the milk to sour in the refrigerator. [2]

Time = ... hours

11

3445U301
07

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